KineticLab_Plan_v2

KineticLab — HackPrinceton Build Plan v2

Stroke & TBI upper-extremity rehabilitation — single-scope refactor

TL;DR

Webcam-only stroke & TBI upper-extremity rehab system. MediaPipe Pose + Hands → Random Forest with ROM-normalized adaptive calibration → React/Three.js exercise scenes via FastAPI WebSocket → Arduino haptics. Ships an automated FMA-UE subscale score (visible items only, ~36/66 points) for the therapist dashboard. $0 hardware floor — sits below FitMi ($349), Saebo ($300–600), Neofect ($99/mo or $15K clinic), and Tyromotion ($75K+). Enroll Healthcare + Hardware tracks; Best Overall is automatic. The unforgettable demo beat is restricted-ROM calibration: a user the system could not read 30 seconds ago now hits every target.

🗺️ Project Timeline — Simplified Flowchart

What changed from v1

Decision	v1	v2
Scope	Stroke rehab + K-12 accessible STEM (dual-market)	Stroke / TBI upper-extremity rehab only
Track enrollment	Healthcare + Education + Hardware	Healthcare + Hardware
Exercises	5 chemistry / physics simulations (H_2O, glucose, etc.)	5 pure rehab tasks mapped to FMA-UE items
Frontend stack	React + Three.js + 3Dmol.js	React + Three.js (3Dmol.js dropped)
Hand grip	Implicit grasp gestures	Out of scope — vision can't measure grip force
Brain visualization	TRIBE v2 live cortical heatmap during rehab	Cut from core build; revisit only if everything ships clean
Clinical advisor	Mentioned	Pre-event email to Prof. Jordan A. Taylor (Princeton IPA Lab)

1. Strategic framing

The differentiation moat

KineticLab competes on three axes that no incumbent stacks together:

- **$0 hardware** — uses any laptop webcam. Every competitor below requires purchased equipment.
- **Adaptive ROM normalization** — a 30-second calibration maps any user's personal range of motion to a 0–1 scale, so a patient with 45° max shoulder flexion can complete the same exercise as a healthy user at 180°.
- **Automated FMA-UE subscale scoring** — every session generates a clinically interpretable score for the therapist dashboard, citing methodology validated to r ≈ 0.99 against clinical raters in published research.

Track enrollment

Track	Status	Prize	Why we win
Best Healthcare Hack	Required, enrolled	Apple Watch Series 11	Strongest clinical case — 795K strokes/yr, validated FMA-UE methodology
Best Hardware Hack	Required, enrolled	BambuLab 3D Printer	Tri-modal Arduino haptics (buzz + vibrate + LED) — creative integration
Best Overall Hack	Automatic	PS5 (×4)	Hits all four rubric axes: Creativity, Utility, Charity, Avidity
Education / Other	Skipped	—	Single-scope discipline; dual-market dilutes the 2-min pitch

2. The clinical case

Numbers that score the Charity criterion. All figures from CDC, AHA, and peer-reviewed sources verified pre-event.

Why this matters

Stat	Source / interpretation
~795,000 US strokes per year	AHA Statistical Update 2025

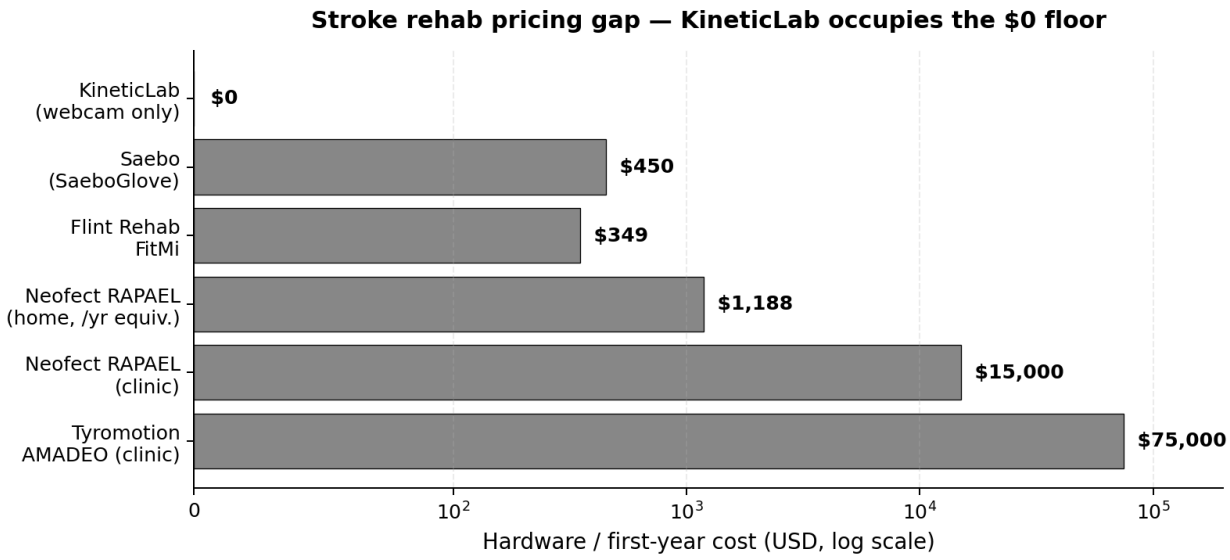
Stat	Source / interpretation
55–80% of acute stroke patients have upper-extremity impairment	55–75% remain impaired in the chronic phase
Only ~31% of stroke survivors access outpatient rehab	CDC MMWR — no US state exceeds 50%
30–50% of those who start rehab quit within 1 year	Post-discharge adherence drops to ~47%
~$56.2B total US stroke cost (2019–2020)	AHA 2025
~2.8M TBIs/year (hospital-based); ~9.76M self-reported (NHIS 2023)	CDC + 2025 NHIS analysis

The home-rehab evidence base

Finding	Citation
Home-based UE intervention more effective than conventional therapy (SMD 0.28, 95% CI 0.12–0.44, $p < 0.001$)	2020 meta-analysis of 26 RCTs
FitMi RCT: gamified home rehab gained 8.0 ± 4.6 FMA-UE points vs 3.0 ± 6.1 conventional	Flint Rehab clinical trial
Game-based UE rehab shows large effect sizes across 48 RCTs / 2,809 participants	2025 meta-analysis (GRADE: low certainty but consistent direction)
Vision-based automated FMA-UE scoring achieves $r = 0.99$ vs clinical raters (MediaPipe + RGBD)	Frontiers in Neuroscience, 2024 — methodological precedent for our scoring module

The pricing gap

KineticLab sits at the $0 hardware floor — every direct competitor requires purchased equipment.



Note: Sword Health acquired Kaia Health for $285M in January 2026 — the market validates camera-based motion analysis for rehabilitation as a high-value category.

3. Technical architecture

Data flow

Webcam → MediaPipe Pose (33 landmarks) + Hands (21/hand) at 25–30 fps → Python feature extraction (joint angles, velocities, normalized positions) → ROM normalization → Random Forest classifier → FastAPI WebSocket → React + Three.js exercise scene. Parallel: Python → pyserial → Arduino (buzzer D6, vibration D8, LED D3). End-to-end latency budget: ~45 ms.

ROM normalization — the technical moat

Three steps, one-time per user, ~30 seconds:

- **1. Baseline capture.** Guide user through max-comfort movements. Track min/max joint angles per landmark.
- **2. Normalize.** Store profile {joint: (min, max)}. Future readings → (current − user_min) / (user_max − user_min). A patient with 45° max shoulder flexion and a healthy user at 180° both map to [0.0, 1.0].
- **3. Adaptive thresholds.** Exercise targets fire at percentages of personal max. Hysteresis (activate 70%, deactivate 50%) prevents flicker. EMA filter on joint

positions; rolling 60-sec variance auto-lowers thresholds when fatigue is detected.

FMA-UE subscale scoring — the credibility piece

The Fugl-Meyer Assessment Upper Extremity (0–66 scale) is the gold standard for post-stroke motor recovery (interrater ICC 0.96). Vision alone cannot measure grip force, so we cap at the visible subscale:

Domain	Points	What we measure
A. Shoulder / elbow / forearm	0–36	Joint angles + reach trajectories from MediaPipe Pose
C. Wrist	0–10	Wrist landmark angles from MediaPipe Hands
E. Coordination / speed	0–6	Bimanual timing + tremor variance
KineticLab subscale total	~52/66	Hand/grip section (D, 0–14) excluded — vision cannot measure grip force

Methodology grounded in Frontiers (2024) r = 0.99, Brain Sciences (2022) r = 0.981, and BIONICS (Zamin et al. 2023) 78–83% item-wise accuracy on 45 acute stroke patients using a single digital camera. Cite these on the Devpost.

4. Rehab exercise library

Five exercises, all pure rehab tasks. Build two to demo polish (Target Reach + Trajectory Trace) — those cover the strongest FMA-UE domains and read instantly to non-clinician judges. Other three rough-in for expo depth.

#	Exercise	What the patient does	FMA-UE items	Demo role
1	Target Reach	Floating targets appear at varying heights/distances; reach with affected arm to extinguish each	A.I–A.IV (shoulder flexion, elbow extension)	Primary demo
2	Trajectory Trace	Trace a moving path on screen with hand; accuracy scored	C.I–C.V (wrist flexion/extension), A.III	Primary demo

#	Exercise	What the patient does	FMA-UE items	Demo role
		against ideal trajectory		
3	Mirror Therapy	Perform movement bilaterally; system mirrors unaffected arm onto affected side in real time	Coordination / speed (E.I–E.III)	Expo depth
4	Forearm Rotation	Hold virtual object; rotate wrist to orient correctly (key-turning metaphor)	B.VII (pronation/supination)	Expo depth
5	Bimanual Reach-and-Place	Pick up virtual object with both hands, transport to target zone	Coordination + bilateral A.I–A.IV	Expo depth

Demo selection rationale

Target Reach + Trajectory Trace covers two of the three FMA-UE domains we score (shoulder/elbow + wrist), uses two distinct gesture vocabularies, and demonstrates ROM scaling visibly. Mirror Therapy is the most clinically distinctive exercise but requires the most polish — keep as expo depth so a 10-minute booth visitor sees breadth without us over-promising on the 2-min stage.

5. Pre-hackathon prep

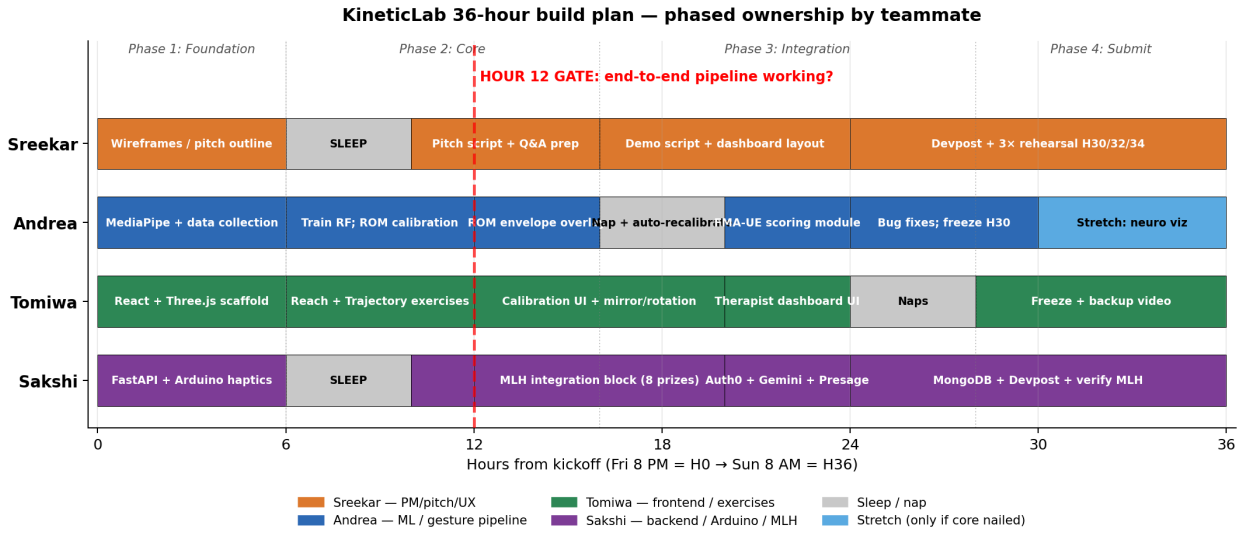
Per-person setup checklist

Person	Pre-event setup
Sreekar	Test webcam framing in target lighting; write pitch outline; pack laptop + charger, HDMI adapter, backup USB webcam, tripod, extension cord, power strip; draft Devpost shell
Andrea	Python 3.10 env with pinned mediapipe, scikit-learn, opencv-python, numpy, fastapi, uvicorn, websockets; verify MediaPipe Pose + Hands on own webcam;

Person	Pre-event setup
	fork kinivi/hand-gesture-recognition-mediapipe (Apache 2.0); read Frontiers (2024) FMA scoring methodology
Tomiwa	Scaffold React + Three.js + @react-three/fiber app; verify Three.js renders smoothly on Ryzen 7 + RTX 5070 Ti; test WebSocket client; review FMA-UE A/C/E item videos on Shirley Ryan AbilityLab site
Sakshi	FastAPI + WebSocket boilerplate; Arduino IDE + Grove buzzer/vibration/LED smoke test; create accounts (Auth0, MongoDB Atlas, ElevenLabs, Gemini); pre-register .tech domain
All	Agree on gesture vocabulary, calibration UX, WebSocket message schema; 30-min integration dry run on Friday afternoon

6. 36-hour build plan

Phased ownership. Each person has one workstream and protected sleep blocks. The Hour 12 Gate is non-negotiable — if the end-to-end pipeline isn't connected by then, all hands fix it before anyone adds new features.



Phase 1 — Foundation (H0–H6, Fri 8 PM – Sat 2 AM)

Person	Tasks
Sreekar	UX wireframes for calibration flow; register .tech domain; GitHub + Devpost shell; pitch outline draft; webcam + tripod setup at venue

Person	Tasks
Andrea	Python env + MediaPipe Pose & Hands; verify 33 body + 21 hand landmarks across all team webcams; collect 50 samples per gesture from each teammate; extract joint angle features
Tomiwa	React + Three.js + @react-three/fiber scaffold; first exercise scene (Target Reach with placeholder targets); WebSocket client with mock data
Sakshi	FastAPI + WebSocket server; Arduino + Grove buzzer/vibration/LED on D6/D8/D3; pyserial bridge; end-to-end smoke test (Python sends “buzz” → Arduino buzzes)

Phase 2 — Core (H6–H12, Sat 2 AM – Sat 8 AM)

Person	Tasks
Sreekar	SLEEP H6–H10 (4 hr). Then pitch script writing + judge Q&A prep
Andrea	Train Random Forest to ≥90% on team data; ROM calibration core: baseline capture, normalization, hysteresis thresholds
Tomiwa	Target Reach exercise (full gesture binding); Trajectory Trace exercise (path generator + accuracy scoring)
Sakshi	SLEEP H6–H10 (4 hr). Then ElevenLabs API integration + pre-generate session narration audio cues

⚠️ HOUR 12 GATE

End-to-end pipeline working? Webcam → MediaPipe → classifier → WebSocket → exercise scene → Arduino haptic. If NO: all hands fix for 2 hours before anyone adds new features. If YES: proceed to Phase 3.

LiveAvatar Expanded Document

🧠 LiveAvatar: Real-Time Rehab Companion

Concept: A HeyGen LiveAvatar serves as the patient-facing interface for stroke/TBI rehabilitation — delivering spoken instructions, adaptive guidance, and positive reinforcement through a real-time human-like presence.

Why a face matters: Stroke/TBI patients frequently have reading deficits, attention impairment, and emotional dysregulation. A calm, responsive human face is more accessible and clinically meaningful than text, audio, or pre-recorded video.

How It Works

The avatar is the **body**. Our LLM is the **brain**.

`Patient speaks/moves → ASR → LLM (clinical prompt layer) → LiveAvatar Lite API → avatar responds`

The avatar delivers step-by-step exercise instructions, monitors patient response, then adapts — simplifying if the patient struggles, celebrating when they succeed. Reinforcement is specific and grounded ("You held that two seconds — up from one last session") rather than generic praise.

Stack

Layer	Tool
Avatar rendering	HeyGen LiveAvatar (Lite/BYOLLM)
Intelligence	GPT-4o or Claude
Voice output	ElevenLabs
Speech input	Whisper / Deepgram

Build Risks

- Confirm LiveAvatar API access on-site Friday
- End-to-end latency target: < 2s
- System prompt must include clinical guardrails

Sreekar's Workstream — Subagent Phase Plan

Why subagents, not agent teams

LiveAvatar and Photon are structurally independent. They share the session data schema (read-only) and the clinical prompt guardrails file. They do not share runtime state during build. Subagents in parallel means no coordination overhead, no inter-agent handoff contracts, and no mid-task blocking.

API access gate (verify at H0)

Before committing to Phase 2, confirm on-site:

- HeyGen LiveAvatar Lite/BYOLLM API access (confirm endpoint is not enterprise-gated)
- Photon Spectrum iMessage API access (confirm Mac server infra or managed infra available)

If either is unavailable → mock-first build. Skeleton clients with swapped-in mock responses. Phase 2 proceeds identically; real API swapped in when access is confirmed.

Phase breakdown

Phase	Hours	Task A (LiveAvatar)	Task B (Photon)
1	H0–H4	HeyGen Lite API client skeleton + mock fallback	Photon iMessage client skeleton + mock fallback
2	H4–H8	Clinical prompt layer + session LLM system prompt	5 message type templates + LLM generator
3	H8–H18	ASR wiring (Whisper/Deepgram → avatar response)	Cron scheduler + session data pull logic
4	H18–H24	E2E test: ASR → LLM → avatar < 2s	E2E test: cron → LLM → iMessage → reply routing

Each phase = one orchestrator call, two parallel subagent Tasks. Review and commit between phases before next batch runs.

.claude/ folder structure

```
.claude/
├── CLAUDE.md
├── skills/
│   ├── liveavatar_api.md
│   ├── photon_spectrum_api.md
│   ├── elevenlabs_streaming.md
│   └── clinical_prompt_guardrails.md
├── agents/
│   └── integration_architecture.md
```

```
└── hooks/
    └── pre-commit
```

Photon Summary

TL;DR: Photon Spectrum lets you send iMessage (and other platform) messages from your backend via API. Here's the condensed Notion-ready writeup for the daily reminder feature.

📲 Photon Spectrum: Daily Recovery Reminders via iMessage

Concept

Stroke and TBI patients have consistently poor adherence to at-home exercise regimens — not from lack of motivation, but because they forget, lose track of time, or have no external nudge. We use **Photon Spectrum** to send personalized, AI-generated daily reminders directly to patients via iMessage — a platform they already use, requiring zero app download or behavior change.

Why iMessage

- No new app to learn or download — critical for this population
- Higher open rates than email or push notifications
- Spectrum delivers messages in under 1 second on Photon's edge network with 99.9% uptime photon
- Supports native platform formatting — polls, buttons, structured responses — so messages feel natural, not robotic photon

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How It Works

```
Scheduler (daily cron)
    → Pull patient session data (last completed exercise, streak, next goal)
    → LLM generates personalized message
    → Photon Spectrum iMessage API sends to patient's phone
    → Patient replies "ready" or taps a button
    → System logs response, optionally launches LiveAvatar session
```

Message Types

Type	Example
Daily reminder	"Good morning [name]. Today's goal: 3 minutes of hand grip. You did it yesterday — let's go."
Streak reinforcement	"Day 5 in a row. That's real consistency. Today: breathing exercises, 5 min."
Missed session nudge	"You skipped yesterday — no problem. Want to do a short 2-minute session now?"
Weekly summary	"This week: 4/7 sessions completed. Your grip strength target improved. Full recap inside."
Quick poll	"How are you feeling today? 💪 Good / 😐 Okay / 😓 Rough" — response routes to easier or standard session

Two-Way Interaction

Photon Spectrum isn't just outbound. Patients can reply:

- "Not today" → system logs, notifies caregiver if pattern repeats
- "Ready" → deeplinks into the app or triggers a LiveAvatar session
- "What's my exercise?" → LLM responds with today's plan inline

This keeps the patient in iMessage — no friction.

Integration with the Broader Stack

Photon sits at the **top of the funnel**. It's the nudge that gets the patient to open the app and engage with the LiveAvatar. The two features are complementary:

`Photon iMessage reminder → patient opens app → LiveAvatar guides session → session logged → next day's Photon message is personalized to outcome`

Build Risk

- Confirm iMessage API access is available on-site (requires Mac server or Photon's managed infra)
- Patient phone number collection/consent needed at onboarding
- Tone guardrails on LLM — messages must never induce guilt or anxiety

Phase 3 — Integration (H12–H24, Sat 8 AM – Sat 8 PM)

Person	Tasks
Sreekar	Demo script v1; rehearse solo; therapist dashboard layout sketches; pitch refinement with team feedback
Andrea	Edge cases (lighting, occlusion); ROM envelope visual overlay on skeleton; user-switch auto-recalibrate; nap H16–H20
Tomiwa	Calibration UI with guided prompts (the demo centerpiece); rough-in Mirror Therapy + Forearm Rotation + Bimanual scenes for expo depth
Sakshi	MLH integration block (8 prizes, parallel workstream): Auth0 (1 hr), Gemini (2 hr), Presage SmartSpectra (3 hr), MongoDB (1 hr), .tech domain (15 min), GoDaddy domain (15 min). Cut anything >3 hr

Phase 4 — Submit (H24–H36, Sat 8 PM – Sun 8 AM)

Person	Tasks	
Sreekar	Devpost writeup; high-quality demo video capture; 3 timed rehearsals at H30, H32, H34	
Andrea	FMA-UE subscale scoring module (the credibility piece). Model freeze H30. Bug fixes only after that. If everything is clean by H30, attempt the neuro-visualization stretch (see Appendix C)	
Tomiwa	Frontend freeze H28; record backup demo video; polish therapist dashboard	
Sakshi	MongoDB session storage; final Devpost submit; verify all 8 MLH prize submissions are checked	

Sleep totals: Sreekar 7h, Sakshi 7h, Andrea ~4h, Tomiwa ~4h. The two heaviest builders take strategic naps; Sreekar must be sharp for the pitch.

7. Two-minute demo script

Setup: laptop on external monitor or projector, webcam positioned for 2m clear space, Arduino visible on table. Sreekar faces camera, Andrea ready for user-switch demo. Target 150 wpm; full script ~380 words.

Time	Beat	Script
0:00–0:15	HOOK	Sreekar reaches toward screen — target extinguishes. Reaches again, sweeps, traces an arc. Buzzer chirps, voice says “Nice reach.”“What you just saw is a stroke rehab

Time	Beat	Script
		session running on nothing but a laptop webcam. No glove, no sensor, no $15,000 clinic equipment. Just a webcam and an open hand.”
0:15–0:45	PROBLEM	“Every year, 795,000 Americans have a stroke. 70% lose use of their arm. Outpatient rehab costs $27,000 per patient — and only 31% of survivors ever access it. Of those who do, a third quit within the first year because the exercises are repetitive and unmotivating. Home-based gamified rehab is proven to match clinic intensity at a fraction of the cost — but the cheapest existing solution still costs $349 in hardware. KineticLab costs zero.”
0:45–1:30	LIVE DEMO	“Watch — I’ll simulate a patient with limited mobility.”Sreekar restricts movement to ~30° shoulder flexion. Red NOT RECOGNIZED indicators. System fails.“The system can’t read me. Now — 30-second calibration.”Guided calibration runs. ROM envelope visibly shrinks. Threshold bars shift live in dashboard.“Now watch.”Same restricted movements → green RECOGNIZED ✓. Targets extinguish in sequence.“Same body, same range. Different exercise — wrist tracing.”Switch to Trajectory Trace. Hand follows path; accuracy meter rises.“And Andrea has a different body —”Andrea steps in. Dashboard shifts, new envelope appears in a different color.“— auto-recalibrates in seconds. Every reach is shoulder flexion. Every trace is wrist extension. The system generates a Fugl-Meyer subscale score for the therapist after every session.”
1:30–1:50	TECH DEPTH	“Under the hood: MediaPipe extracts 33 body and 21 hand landmarks per frame. A Random Forest classifier with per-user ROM-normalized thresholds drives the exercise logic through a FastAPI WebSocket. Arduino delivers tri-modal haptic feedback. Our scoring methodology is benchmarked against published vision-based FMA-UE work — Frontiers 2024 reports r = 0.99 against clinical raters using the same MediaPipe pipeline. Runs on any laptop. Calibrates to any new user in 30 seconds.”
1:50–2:00	CALL TO ACTION	“Six hundred thousand stroke survivors lose arm function every year. The cheapest existing system costs $349. Ours costs zero, runs on what they already own, and generates clinically interpretable scores. KineticLab is rehabilitation

Time	Beat	Script
		that meets people where they are — at home, with a webcam, and a hand they're learning to use again."

Contingencies

- Webcam glitches → switch to phone-as-webcam via Camo or EpocCam (rehearse this).
- Gesture misreads → "like any ML system, sometimes it needs a second try" → repeat the gesture.
- Calibration too slow → pre-load a saved profile.
- Live demo fails → narrate over backup video recorded H32–H34.

8. Risk register

Risk	L × I	Mitigation	Trigger to act
Webcam quality / lighting poor	M × M	Test at venue H0; bring 2–3 webcams (MLH supplies 8 Logitechs); secondary laptop as backup; phone-as-webcam via Camo/EpocCam	H0: landmark detection flaky
MediaPipe dependency conflict	L × H	Pin exact versions in requirements.txt; team uses same Python (3.10); test install on fresh env pre-event	H1: pip install fails on any laptop
Gesture accuracy < 70%	M × H	Reduce vocabulary; collect more samples; RF → SVM fallback; rule-based on joint angles	H8: <75% on team
Latency > 200 ms	L × H	Everything on localhost; lightweight models; pre-compute features; reduce MediaPipe complexity tier	H10: >150 ms
Background / other people interfere	M × M	Position camera wall-facing; MediaPipe primary-person tracking handles most cases; rehearse in clutter	Always
Venue Wi-Fi unreliable	M × M	Core runs locally; pre-cache ElevenLabs + Gemini responses;	On arrival

Risk	L × I	Mitigation	Trigger to act
		local JSON for Mongo	
FMA-UE scoring overclaim	L × H	Cite published methodology explicitly; do NOT call it FDA-validated; frame as research-grade subscale proxy	Pitch rehearsal
Demo-day failure	M × Crit	Record high-quality backup video H32–H34; narrate over video if live fails	Always
Team illness / exhaustion	M × H	Enforce sleep plan; written pitch so anyone can deliver	Continuous
MLH integration breaks core	M × M	All MLH is additive, never on critical path; cut any integration >3 hr	Continuous

9. Strategic recommendations

Do these three things

- **End-to-end pipeline by H12.** A judge should be able to wave at the camera and see a target extinguish. Doesn't have to be polished — has to be connected.
- **Adaptive calibration is the demo centerpiece.** In a field of 50+ projects, the unforgettable moment is: restricted-ROM user fails → 30-sec calibration → succeeds visibly. Dedicate ≥4 hours to making this flawless.
- **Rehearse with a timer 3×.** Two minutes is brutal. Most teams improvise and run over. H30, H32, H34.

Don't do these three things

- **Don't add hardware that doesn't earn its seat.** Webcam + Arduino haptics is the surface area. Adding depth sensors or extra peripherals adds risk without strengthening the pitch — the accessibility narrative requires minimal hardware.
- **Don't overclaim on clinical validity.** FMA-UE subscale scoring is research-grade methodology, not an FDA-cleared diagnostic. Frame it accurately. Judges with healthcare backgrounds will catch overclaims and penalize them.
- **Don't spread MLH integrations across the team.** One person (Sakshi), one block (H12–H20). Cut any integration >3 hr.

If everything ships clean — stretch ranking

In priority order, only if Phase 4 is on track by H30:

#	Stretch	Rationale
1	Therapist analytics dashboard (ROM trends + session-over-session FMA-UE delta)	Highest pitch ROI — turns the system from “a tool” into “a longitudinal care platform”
2	Open-source neuroimaging visualization	Use HCP or Neurosynth motor-task cortical maps as pre-rendered “motor cortex regions engaged” imagery on the dashboard. Honest, no live inference, drives home neurological narrative without TRIBE v2 risk
3	Add Mirror Therapy as a third demo exercise	FMA-UE coordination domain coverage; clinically distinctive
4	TRIBE v2 (post-hoc visual/audio cortex predictions)	Only if cloud GPU is set up and a teammate has time. Reframe as “cortical response to rehab game stimuli,” NOT “motor cortex during exercise.” See Appendix C

Appendix A — FMA-UE quick reference

Useful for the pitch, judge Q&A, and the Devpost write-up. Source: Shirley Ryan AbilityLab RehabMeasures Database; Fugl-Meyer et al. 1975.

Domain	Points	Items / our coverage
A. Upper extremity (shoulder/elbow/forearm)	0–36	Reflexes, flexor/extensor synergy, mixed synergy, movement out of synergy. KineticLab covers via Target Reach
B. Wrist	0–10	Stability with elbow flexed/extended, flexion/extension, circumduction. KineticLab covers via Trajectory Trace + Forearm Rotation
C. Hand	0–14	Mass flexion/extension, grasp variants, pincer. NOT COVERED — vision cannot measure grip force
D. Coordination / speed	0–6	Finger-to-nose tremor, dysmetria, time. KineticLab partially covers via Bimanual

Domain	Points	Items / our coverage
		Reach-and-Place
Total	0–66	KineticLab subscale score: 0–52 (excludes hand grip)

Severity classifications: severe 0–19, moderate 20–47, mild 48–66. Minimal clinically important difference: 4.25–7.25 points (chronic, mild-moderate); 9–13 points (subacute, moderate-severe). Interrater reliability ICC = 0.96.

Appendix B — Competitor cheat sheet

Product	Hardware	Cost	How KineticLab beats them
Tyromotion AMADEO	Robotic finger device, clinic-only	~$50K–$100K+ institutional	Consumer-accessible; no hardware purchase
MindMaze MindMotion	Markerless optical capture, FDA-cleared	Institutional licensing	Same modality (camera) but consumer-grade and free
Neofect RAPAEL	Smart glove with IMU + flex sensors	$99/mo home or ~$15K clinic	No glove required; works with whatever ROM the patient has
Jintronix	Depth-sensor-based	Discontinued (Kinect)	MediaPipe replaces Kinect; runs on any laptop
Saebo (SaeboGlove)	Mechanical orthosis	~$300–$600	Software-only; we measure what they only assist
Flint Rehab FitMi	2 wireless sensor pucks, FDA-listed	$349	$0 hardware; same gamification model with adaptive ROM
Kaia Health (Sword)	Smartphone camera CV (MSK pain focus)	Acquired for $285M (1/2026)	Stroke-specific; explicitly trained for hemiparetic ROM
MIRA Rehab	Depth-sensor exergames, CE-marked	Institutional	Webcam, not depth sensor — works on devices people already own

Appendix C — Stretch: TRIBE v2 (only if everything ships)

Read this before touching TRIBE v2

TRIBE v2 predicts how the brain responds to incoming stimuli (video, audio, text) — it does NOT predict motor cortex activation from performing movements. Using it to show “brain activation during rehab” is scientifically incorrect. Reframe required: present TRIBE v2 as predicting cortical response to the visual/audio stimulation of the rehab game, NOT motor cortex during exercise. Anything else is overclaiming and a healthcare-savvy judge will catch it.

Hard requirements

- Cloud A100 (40 GB VRAM minimum). RTX 5070 Ti has 16 GB — insufficient. Budget ~$1–2/hr on Lambda/RunPod/Modal.
- LLaMA 3.2-3B HuggingFace access token (gated model — accept license before event).
- ~22 GB model download. Cache to cloud volume before hackathon if possible.
- Owner: Andrea, only after H30 if model is frozen and core demo is solid.

Minimum viable path

#	Step	Time
1	Spin up cloud GPU instance (≥40 GB VRAM)	30 min
2	Install tribev2, accept LLaMA license, download encoders	60 min
3	Record 30–60s rehab session video (.mp4)	5 min
4	Run model.predict(events=df) → activation tensor (T, 20484)	10 min
5	nilearn view_img_on_surf() → interactive HTML brain map	60 min
6	Embed HTML iframe in therapist dashboard “Researcher View” tab	60 min
	Total realistic build time	≥4 hours, fragile

Better stretch alternative — open-source motor cortex maps

Faster, lower risk, scientifically defensible: pre-render motor cortex region images from open-source neuroimaging atlases (Neurosynth, HCP motor task) and overlay them on the therapist dashboard with annotations like “primary motor cortex (M1) — engaged

during shoulder flexion exercise.” No live inference, no cloud GPU, no $300 LLaMA download. Same neurological narrative impact, finished in 2–3 hours instead of 8+. Recommended over TRIBE v2 for this hackathon.

🗂️ .claude/ — Sreekar's Workstream Config

⌨️ Claude Code Prompt Sequence — Sreekar's Workstream